

CHARLES BENELLO

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EDUCATION

The University of Chicago

Chicago, IL

Master of Science in Financial Mathematics

Expected Mar 2027

- Wayne Booth Graduate Student Teaching Prize (2026) - UChicago's top graduate teaching award
- Coursework: Distributed Systems, ML in SWE, Agentic AI, Reinforcement Learning, Computing for Finance

The University of Chicago | GPA: 3.75/4.0

Chicago, IL

Bachelor of Science in Mathematics and Computer Science (Systems), Minor in Korean

Jun 2025

- Coursework: Database Systems, Operating Systems, Algorithms, Concurrency Control, Computer Architecture, Networks

EXPERIENCE

Database Research Assistant

Apr 2023 – Present

ChiData Research Lab, University of Chicago

Chicago, IL

- First-authored VLDB 2026 paper (CrocSort) by designing a parallel external merge sort in Rust that completes under tight memory budgets where PostgreSQL, DuckDB, and ClickHouse abort, jointly planning memory and phase-specific thread counts.
- Co-authored CIDR 2025 paper on resource-adaptive query execution with paged memory management, enabling query operators to adapt to available memory at runtime under contention.
- Sole-authored ADMS 2026 paper on a lightweight prediction model for sort configuration on cloud and disaggregated hardware, transferring calibration across machines within 3–5% of optimal without brute-force sweeps.

GPU Systems Researcher

Jun – Sep 2025

EPFL (DIAS Lab, Prof. Anastasia Ailamaki)

Lausanne, Switzerland

- Engineered a macroblock-based GPU decompression strategy in CUDA/C++ that cut out-of-memory query times by 50–200% on large analytical workloads.
- Built cost-based simulation models predicting query performance across workload complexities, enabling adaptive decompression strategy selection at plan time.
- Designed join patterns for TPC-DS benchmarks with resource contention models across I/O, decompression, and compute pipelines.
- Benchmarked GPU compression trade-offs against CPU baselines, shaping a new cost-based research direction with a forthcoming publication.

Software Engineering Intern

Feb 2026 – Present

Alter Domus

Chicago, IL

- Lifted document extraction accuracy from 50% to 92–97% by architecting a multi-stage pipeline with schema-enforced structured output and cross-reference verification for 100+ page financial filings.
- Built an LLM-as-a-Judge evaluation framework with automated benchmarks for faithfulness, numerical accuracy, and completeness; cut LLM inference costs by optimizing context handling.

Quantitative Researcher

Mar 2026 – Present

PayPal (UChicago Project Lab)

Chicago, IL

- Built a self-supervised transaction foundation model (TabBERT-style hierarchical 2-level transformer, ~5M params) pretrained on 14.5M unlabeled transactions, reaching AUC 0.974 / F1 0.822 on downstream fraud detection at 0.12% positive rate.
- Ran 87+ ablation experiments on UChicago's Midway3 SLURM cluster across architecture, tokenization, time encoding, masking rate, and sequence length; implemented per-field masking and SPRM from PANTHER (NeurIPS 2025) and PRAGMA (Revolut 2026), with per-field masking alone yielding +1.9% AUC and +16.8% F1.

PROJECTS

GPU CrocSort | CUDA, C++ | github.com/cmбенello/gpu-research

2025–Present

- Ported CrocSort (VLDB 2026) to GPU. On TPC-H SF50 ORDER BY lineitem (300M rows, 36 GB) beats DuckDB by 20.1× and Polars by 34.5× on a single PCIe-Gen3 GPU; 2.4× faster than MendSort (best published single-node CPU sort, JouleSort 2023). Designed a cost-based plan-time selector between compact-key and full-key paths with a measured 32-byte entropy crossover; cut PCIe amplification to 0.14× via key-only upload + permutation gather + triple-buffered streaming.

Concurrent Database Engine | Rust | github.com/cmбенello/CrustyDB

2025

- Built a multi-threaded relational database from scratch in Rust: B+ tree indexing, concurrent buffer pool with page-level locking, heap storage, and a SQL query execution engine.

Transit Coverage Optimizer | Python | cmбенello.github.io/covtsp

2026

- Beat the Guinness World Record by 60 minutes (16h45m vs. 17h45m) with an automated Covering TSP pipeline: fetches GTFS feeds and Google Maps walking data, builds a time-expanded graph (160K nodes, 1.26M edges), and runs k-NN with urgency scoring and epsilon-greedy search. Generalized to NYC (475 stations) and Berlin.

TECHNICAL SKILLS

Languages: C/C++, Rust, CUDA, Python, SQL, Go, TypeScript, JavaScript

Tools: PostgreSQL, DuckDB, Databricks, Docker, Git, Linux, PyTorch, Next.js, FastAPI

Concepts: Database Internals, Query Execution & Optimization, External Sort, Distributed Systems, GPU Programming, Concurrency, Cost-Based Optimization, Performance Engineering

PUBLICATIONS

[1] R. Otaki*, C. Benello*, et al. "CrocSort: Resource-Efficient, Skew-Resilient Parallel External Merge Sort." *VLDB 2026*. (*Equal contribution)

[2] C. Benello. "Getting the Most Out of External Sorting in Cloud and Disaggregated Environments." *ADMS 2026*.

[3] R. Otaki, J.H. Chang, C. Benello, A. Elmore, G. Graefe. "Resource-Adaptive Query Execution with Paged Memory Management." *CIDR 2025*.

ADDITIONAL

Languages: Korean, Italian (Intermediate); French (Basic) | **Interests:** Running (Chicago Marathon '22, '23, '24), Climbing, Weightlifting